

Deepak Sathyan

Curriculum Vitae

[arXiv](#) | [InspireHEP](#) | [Google Scholar](#) | [ORCID](#) | [Website](#)

✉ dsathyan@tamu.edu

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX
Sept 2024 – Aug 2027
3 years

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

College Park, MD
Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Boston, MA
Aug 2014 – May 2018

Research Interests

Beyond the Standard Model searches with Collider Physics: LHC and muon collider phenomenology, Randall-Sundrum and baryon-number-violation models

Searches for sub-GeV Dark Matter: dark matter-neutrino interactions, dark photon and dark sector production, cosmic ray-dark matter scattering

Neutrino Physics: supernova neutrino attenuation and detection, open-system observables, neutrinophilic mediators, neutrino NSI

Publications

5. New laboratory constraints on neutrinophilic mediators

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

July 2024

4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

Apr 2024

3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
 K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)
1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020
 K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Preprints

4. **Intensity-Frontier Signals of Warped Extra Dimensions** Aug 2026
 D. Kim, D. Sathyan, A. Verma
arxiv.org/abs/2608.20482
3. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026
 B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010
2. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
 A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064
1. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000

Articles in Preparation

1. **LHC as a Beam Dump: Probing Light Dark Photons**
 B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
 (In preparation)
2. **New Probes of Dark Sectors at a Muon Collider**
 B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
 (In preparation)
3. **Exploring open systems approach to neutrino oscillations**
 A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
 (In preparation)

4. LHC Signals for KK Gravitons Producing 8-Particle Final States

K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan

(In preparation)

Snowmass2021 Contributions

5. TF07 Snowmass Report: Theory of Collider Phenomena

Oct 2022

F. Maltoni et al.

arxiv.org/abs/2210.02591

4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021

Sept 2022

K. Agashe et al.

arxiv.org/abs/2209.11267

3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021

Sept 2022

T. Bose et al.

arxiv.org/abs/2209.13128

2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond

Apr 2022

K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan

arxiv.org/abs/2204.02928

1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models

Mar 2022

K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan

arxiv.org/abs/2203.13305

Muon g-2 Articles

3. The straw tracking detector for the Fermilab Muon g-2 Experiment

Nov 2021

B.T. King et al.

arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)

2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab

Apr 2021

Muon g-2 Collaboration

arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm

Apr 2021

Muon g-2 Collaboration

arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Ph.D. Thesis

- 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Teaching

- Principles of Modern Physics** Spring 2024
Teaching Assistant, University of Maryland
- Electricity and Magnetism I** Spring 2024
Teaching Assistant, University of Maryland
- Mathematical Methods for Physics II** Fall 2021, Fall 2022
Teaching Assistant, University of Maryland
- Fundamentals of Physics I** Summer 2019, Spring 2022
Teaching Assistant, University of Maryland
- Introduction to Quantum Mechanics I** Fall 2020
Teaching Assistant, University of Maryland
- Fundamentals of Physics for Biologists II** Spring 2019, Spring 2020
Teaching Assistant, University of Maryland
- Fundamentals of Physics for Biologists** Fall 2018, Fall 2019
Teaching Assistant, University of Maryland
- [FILL IN: guest lecture title/venue] [FILL IN: date]
Guest Lecture — [FILL IN: course and institution]
- [FILL IN: TAMU teaching activity] [FILL IN: date]
[FILL IN: role and details]

Seminars

- 9. HEP Seminar at Northwestern University** Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 8. Fermilab Cosmic Physics Center Seminar** Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 7. CMS Exotica general meeting** May 2024
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
- 6. Seminar at Texas A&M** Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

5. Seminar at UT Austin	Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	
4. CMS SUSY general meeting	Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	
3. HEP Seminar at University of Pittsburgh	Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	
2. HEP Seminar at Northwestern University	Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	
1. Theory Seminar at Washington University in St. Louis	Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	

Conference Talks

14. Light Dark World 2026 at Carleton University [Invited]	July 2026
Can the LHC be sensitive to the Light Dark World?	
13. CETUP* Workshop [Invited]	June 2026
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering	
12. Particle Physics on the Plains at the University of Kansas [Contributed]	Nov 2025
A Baryon and Lepton Number Violation Model Testable at the LHC	
11. CETUP* Workshop [Invited]	June 2025
Can the LHC be sensitive to light dark mediators?	
10. Phenomenology Conference at the University of Pittsburgh [Contributed]	May 2025
Can the LHC be sensitive to light dark mediators?	
9. Particle Physics on the Plains at the University of Kansas [Contributed]	Nov 2024
A comprehensive analysis of supernova neutrino-dark matter interactions	
8. Texas TACOS @ UT Austin [Invited]	Oct 2024
A comprehensive analysis of supernova neutrino-dark matter interactions	
7. NuFact 2024 Conference at Argonne National Laboratory [Invited]	Sept 2024
A comprehensive analysis of supernova neutrino-dark matter interactions	
6. Mitchell Conference at Texas A&M University [Invited]	May 2024
A comprehensive analysis of supernova neutrino-dark matter interactions	

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| 5. Particle Physics on the Plains at the University of Kansas [Contributed]
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Oct 2023 |
| 4. Phenomenology Conference at the University of Pittsburgh [Contributed]
Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos | May 2023 |
| 3. Phenomenology Conference at the University of Pittsburgh [Contributed]
Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I) | May 2022 |
| 2. April APS Meeting [Contributed]
Signals of KK graviton from extended warped extra dimensions at the LHC (II) | Apr 2021 |
| 1. Phenomenology Conference at the University of Pittsburgh [Contributed]
Signals of KK graviton from extended warped extra dimensions at the LHC (II) | May 2020 |

Service

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Conference Organizer — Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Conference Organizer — Texas A&M University, May 13-16, 2025

Mitchell Institute High Energy Phenomenology, Experiment, and Cosmology Seminar Series
Weekly Seminar Organizer - Texas A&M University, Fall 2024 - Present

Awards

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| Breakthrough Prize in Fundamental Physics
Awarded to the Muon g-2 Collaboration; recognized as a collaboration member | Apr 2026 |
| Teaching Assistant Award
University of Maryland Department of Physics | Dec 2019 |

Schools Attended

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| GGI Lectures on the Theory of Fundamental Interactions | Jan 2023 |
| TASI 2024: The Frontiers of Particle Theory | June 2024 |

Outreach

BU Student Mentorship Program PRISM (2016-2018)
Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

BU Outreach for Local High Schools

Helped demonstrate measurement of the charge-to-mass ratio of the electron for local high schoolers in physics

TAMU Physics and Engineering Festival (2025-present)

Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" (December 2024)

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules

References

Dr. Kaustubh Agashe

University of Maryland, College Park — kagashe@umd.edu

Dr. Bhaskar Dutta

Texas A&M University — dutta@tamu.edu

Dr. Jason Kumar

University of Hawai'i — jkumar@hawaii.edu

Dr. Bhupal Dev

Washington University in St. Louis — bdev@wustl.edu